Overlap-continuous Floquet branches

$\varphi = 0$ $\varphi = \pi/4$ $\varphi = \pi/2$

(a) (b) (c)

Unwrapped quasienergy $\tilde{\varepsilon}_n/t_0$

(d) (e) (f)

State dimension D_n

(g) (h) (i)

Assigned-neighbor overlap $|\langle\psi_i|\psi_{i+1}\rangle|^2$

Drive frequency ω/t_0

Branch 1
Branch 2
Branch 3
Branch 4
Fixed-reference outer-set mean
Dynamic outer-set mean
Tracking / near-gap warning
Overlap = 0.8
Unresolved connection interval

$L = 89$, $\alpha = 55/89$, $t_0 = 1$, $\Delta_0 = 0.5$, $V = 2.5$, $W = 0.01$, $A = B = 0.8$, $\theta = 0$, PBC
Gauge anchored at $\omega/t_0 = 8$; four positive-q branches ordered by their anchor q. Assignment compares neighbors toward the anchor.
Crosses: low overlap (< 0.8), ambiguity, or near gaps. Shaded: unresolved branch-connection intervals (any tracked state).
These tracks are not guaranteed adiabatic branches; low-confidence changes do not establish a transition.